

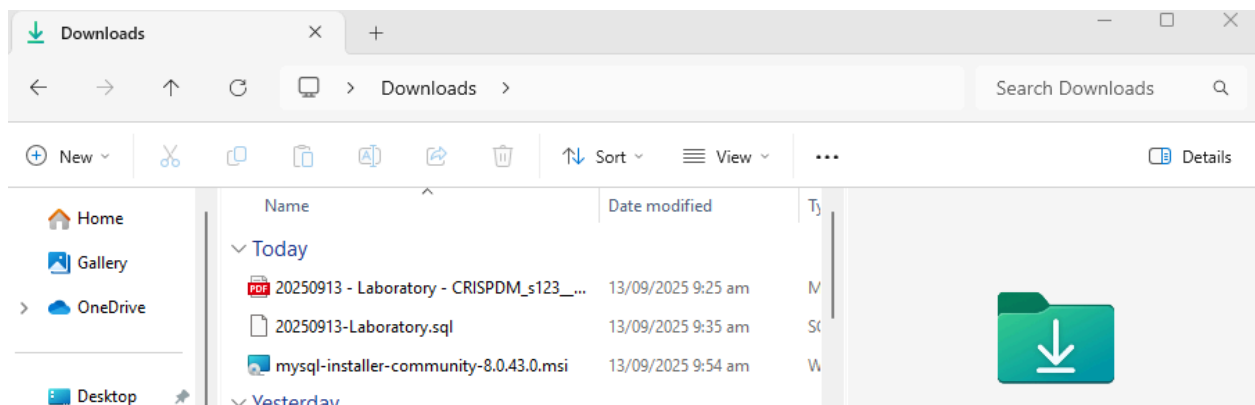
|                                  |                          |
|----------------------------------|--------------------------|
| Name: Sarmineto, Sean Tristan R. | Date: September 13, 2025 |
| Section: CS41S1                  |                          |

### Lab Activity-03: CRISP-DM | Exploring CRISP-DM Phases 1, 2, & 3 using Bank Marketing Data Total Points: 65

CRISP-DM: Step-1 | Business Understanding OBJECTIVE: This activity is designed to explore the initial phases of the CRISP-DM framework – Business Understanding, Data, Understanding, and Data Preparation – through scenario-based analysis. It involves identifying real world problems in the banking sector, examining a publicly available dataset, and preparing the data for future modeling. The exercise emphasizes the importance of aligning data mining efforts with clearly defined business goals, resource considerations, and contextual constraints, forming a foundation for informed decision-making and effective analytical planning.

CRISP-DM: Step-2 | Data Understanding INSTRUCTIONS:

1.) Download the MySQL data-dump located at [https://github.com/AI-Artisans/bda\\_cs41s1/tree/main/20250913-Lab-Wk8](https://github.com/AI-Artisans/bda_cs41s1/tree/main/20250913-Lab-Wk8)



2.) From the command line (cmd.exe), restore the MySQL data-dump using the command below:

```
mysql -u [MySQL DB User] [MySQL database] < [local path of MySQL data-dump file]
```

example:

mysql -u root testdb < D:\20250913-Laboratory.sql

The screenshot displays the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' and 'PERFORMANCE' sections. The main window shows a query result for the 'bank\_marketing' table. The query is 'SELECT \* FROM bank\_marketing'. The result set is displayed in a table with columns: id, age, job, marital, education, default\_status, balance, housing, loan, contact, day, month, duration, campaign, pdays, previous, outcome, and target. The table contains 25 rows of data. The bottom status bar shows the system temperature as 28°C and the time as 10:45:02 am on 13/09/2025.

MySQL Workbench

LabActivity - Ylming - not su... x

File Edit View Query Database Server Tools Scripting Help

Navigator

Query 1 Administration - Data Import/Export

1 SELECT \* FROM bank\_marketing

Limit to 1000 rows

Result Grid Filter Rows

| id | age | job          | marital  | education | default_status | balance | housing | loan | contact | day | month | duration | campaign | pdays | previous | outcome | target |
|----|-----|--------------|----------|-----------|----------------|---------|---------|------|---------|-----|-------|----------|----------|-------|----------|---------|--------|
| 1  | 58  | management   | married  | tertiary  | no             | 2143.00 | yes     | no   | unknown | 5   | may   | 261      | 1        | -1    | 0        | unknown | no     |
| 2  | 44  | technician   | single   | secondary | no             | 29.00   | yes     | yes  | unknown | 5   | may   | 151      | 1        | -1    | 0        | unknown | no     |
| 3  | 33  | entrepreneur | married  | secondary | no             | 2.00    | yes     | yes  | unknown | 5   | may   | 76       | 1        | -1    | 0        | unknown | no     |
| 4  | 47  | blue-collar  | married  | unknown   | no             | 1506.00 | yes     | no   | unknown | 5   | may   | 92       | 1        | -1    | 0        | unknown | no     |
| 5  | 33  | unknown      | single   | unknown   | no             | 1.00    | no      | no   | unknown | 5   | may   | 198      | 1        | -1    | 0        | unknown | no     |
| 6  | 35  | management   | married  | tertiary  | no             | 231.00  | yes     | no   | unknown | 5   | may   | 139      | 1        | -1    | 0        | unknown | no     |
| 7  | 28  | management   | single   | tertiary  | no             | 447.00  | yes     | yes  | unknown | 5   | may   | 217      | 1        | -1    | 0        | unknown | no     |
| 8  | 42  | entrepreneur | divorced | tertiary  | yes            | 2.00    | yes     | no   | unknown | 5   | may   | 380      | 1        | -1    | 0        | unknown | no     |
| 9  | 58  | retired      | married  | primary   | no             | 121.00  | yes     | no   | unknown | 5   | may   | 90       | 1        | -1    | 0        | unknown | no     |
| 10 | 40  | technician   | single   | secondary | no             | 993.00  | yes     | no   | unknown | 5   | may   | 55       | 1        | -1    | 0        | unknown | no     |
| 11 | 41  | admin.       | divorced | secondary | no             | 270.00  | yes     | no   | unknown | 5   | may   | 222      | 1        | -1    | 0        | unknown | no     |
| 12 | 29  | admin.       | single   | secondary | no             | 390.00  | yes     | no   | unknown | 5   | may   | 137      | 1        | -1    | 0        | unknown | no     |
| 13 | 53  | technician   | married  | secondary | no             | 6.00    | yes     | no   | unknown | 5   | may   | 517      | 1        | -1    | 0        | unknown | no     |
| 14 | 58  | technician   | married  | unknown   | no             | 71.00   | yes     | no   | unknown | 5   | may   | 71       | 1        | -1    | 0        | unknown | no     |
| 15 | 57  | services     | married  | secondary | no             | 162.00  | yes     | no   | unknown | 5   | may   | 174      | 1        | -1    | 0        | unknown | no     |
| 16 | 51  | retired      | married  | primary   | no             | 229.00  | yes     | no   | unknown | 5   | may   | 353      | 1        | -1    | 0        | unknown | no     |
| 17 | 40  | admin.       | single   | unknown   | no             | 13.00   | yes     | no   | unknown | 5   | may   | 98       | 1        | -1    | 0        | unknown | no     |
| 18 | 57  | blue-collar  | married  | primary   | no             | 52.00   | yes     | no   | unknown | 5   | may   | 38       | 1        | -1    | 0        | unknown | no     |
| 19 | 40  | retired      | married  | primary   | no             | 60.00   | yes     | no   | unknown | 5   | may   | 219      | 1        | -1    | 0        | unknown | no     |
| 20 | 33  | services     | married  | secondary | no             | 0.00    | yes     | no   | unknown | 5   | may   | 54       | 1        | -1    | 0        | unknown | no     |
| 21 | 28  | blue-collar  | married  | secondary | no             | 723.00  | yes     | yes  | unknown | 5   | may   | 262      | 1        | -1    | 0        | unknown | no     |
| 22 | 56  | management   | married  | tertiary  | no             | 779.00  | yes     | no   | unknown | 5   | may   | 164      | 1        | -1    | 0        | unknown | no     |
| 23 | 32  | blue-collar  | single   | primary   | no             | 23.00   | yes     | yes  | unknown | 5   | may   | 160      | 1        | -1    | 0        | unknown | no     |
| 24 | 25  | services     | married  | secondary | no             | 50.00   | yes     | no   | unknown | 5   | may   | 342      | 1        | -1    | 0        | unknown | no     |
| 25 | 46  | retired      | married  | primary   | no             | 0.00    | yes     | yes  | unknown | 5   | may   | 181      | 1        | -1    | 0        | unknown | no     |

bank\_marketing 3 x

Output

| # | Time     | Action                                     | Message              | Duration / Fetch      |
|---|----------|--------------------------------------------|----------------------|-----------------------|
| 1 | 10:37:35 | USE testdb                                 | 0 row(s) affected    | 0.000 sec             |
| 2 | 10:37:44 | SHOW TABLES                                | 1 row(s) returned    | 0.000 sec / 0.000 sec |
| 3 | 10:38:00 | DESCRIBE bank_marketing                    | 18 row(s) returned   | 0.000 sec / 0.000 sec |
| 4 | 10:38:28 | SELECT * FROM bank_marketing LIMIT 0, 1000 | 1000 row(s) returned | 0.000 sec / 0.000 sec |

Object Info Session

28°C Partly sunny

Search

ENG US 10:45:02 am 13/09/2025

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

REFLECTION: 1.) What challenges did you experience during the MySQL data-dump restoration? (5 pts)

- The Challenges that I have faced when doing the data-dump restoration was making the dataset appear, because I was confused at first on how I should start or tackle this activity.

2.) With respect to the newly created MySQL table, complete the details below (5 pts):

|                             |               |
|-----------------------------|---------------|
| <b>Number of Attributes</b> | <b>18</b>     |
| <b>Number of Objects</b>    | <b>45,211</b> |

3.) With respect to the objects inside the table, complete the details below (10 pts):

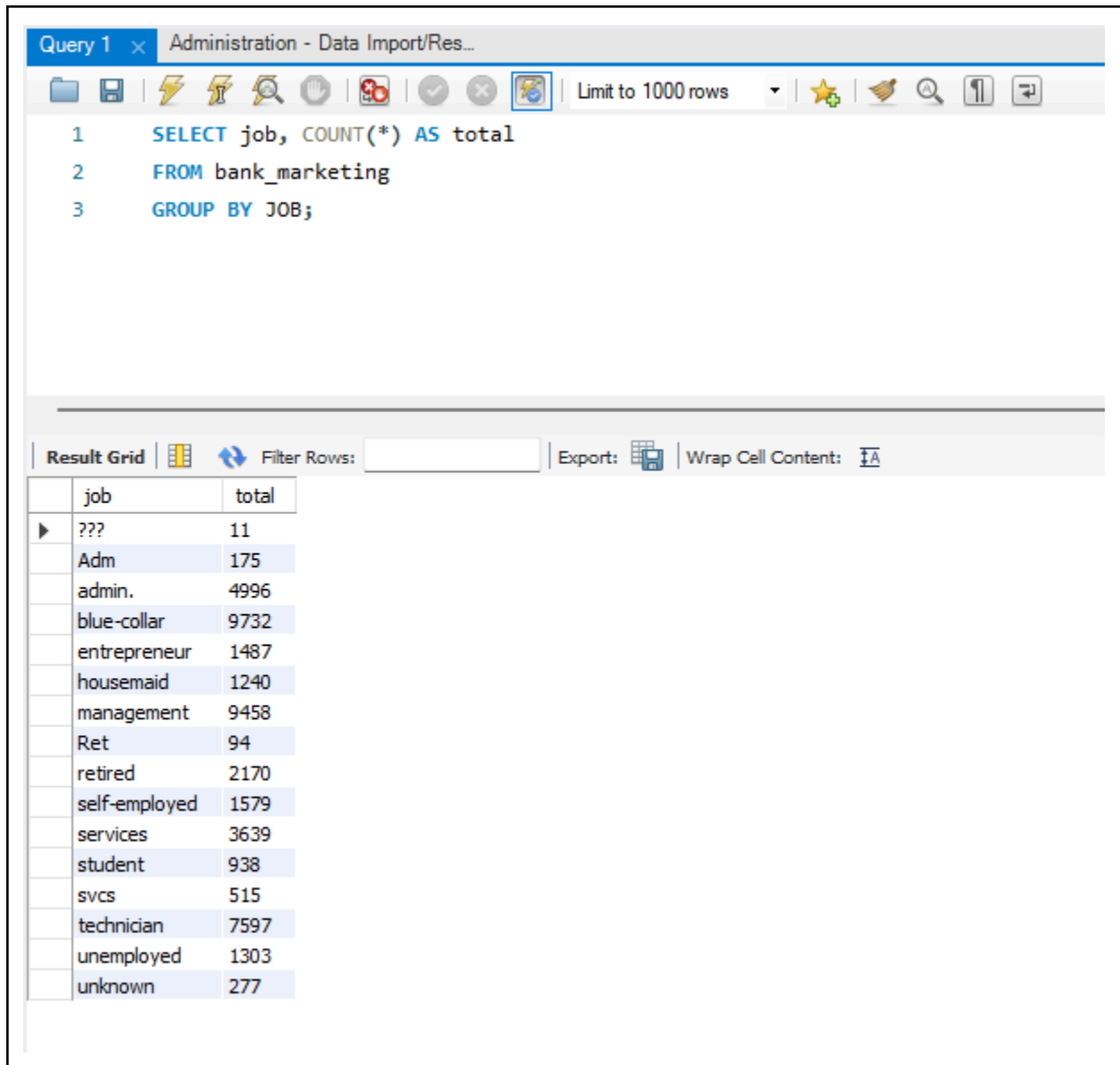
| <b>Attributes</b> | <b>Data Type</b> |
|-------------------|------------------|
| ID                | Integer          |
| AGE               | Integer          |
| JOB               | Varchar/Text     |
| MARITAL           | Varchar/Text     |
| EDUCATION         | Varchar/Text     |
| DEFAULT_STATUS    | Varchar/Text     |
| BALANCE           | Decimal          |
| HOUSING           | Varchar/Text     |
| LOAN              | Varchar/Text     |
| CONTACT           | Varchar/Text     |
| DAY               | Integer          |
| MONTH             | Varchar/Text     |

|          |              |
|----------|--------------|
| DURATION | Integer      |
| CAMPAIGN | Integer      |
| PDAYS    | Integer      |
| PREVIOUS | Integer      |
| OUTCOME  | Varchar/Text |
| TARGET   | Varchar/Text |

## CRISP-DM: Step-3 | Data Preparation

INSTRUCTIONS:

1.) Perform initial data exploration using the restored data-dump. Complete item #1 in Reflection section



The screenshot shows a SQL query editor window titled "Query 1" with the following SQL code:

```
1 SELECT job, COUNT(*) AS total
2 FROM bank_marketing
3 GROUP BY JOB;
```

Below the query editor, the "Result Grid" tab is active, displaying the results of the query. The table has two columns: "job" and "total". The results are as follows:

| job           | total |
|---------------|-------|
| ???           | 11    |
| Adm           | 175   |
| admin.        | 4996  |
| blue-collar   | 9732  |
| entrepreneur  | 1487  |
| housemaid     | 1240  |
| management    | 9458  |
| Ret           | 94    |
| retired       | 2170  |
| self-employed | 1579  |
| services      | 3639  |
| student       | 938   |
| svcs          | 515   |
| technician    | 7597  |
| unemployed    | 1303  |
| unknown       | 277   |

Query 1 x Administration - Data Import/Res...

Limit to 1000 rows

```
1 SELECT marital, COUNT(*) AS total
2 FROM bank_marketing
3 GROUP BY marital;
```

Result Grid

Filter Rows:  Export: Wrap Cell Content:

|   | marital  | total |
|---|----------|-------|
| ▶ | Div      | 583   |
|   | divorced | 4624  |
|   | married  | 27214 |
|   | single   | 12790 |

Query 1 x Administration - Data Import/Res...

Limit to 1000 rows

```
1 SELECT education, COUNT(*) AS total
2 FROM bank_marketing
3 GROUP BY education;
```

Result Grid

Filter Rows:  Export: Wrap Cell Content:

|   | marital  | total |
|---|----------|-------|
| ▶ | Div      | 583   |
|   | divorced | 4624  |
|   | married  | 27214 |
|   | single   | 12790 |

Query 1 x Administration - Data Import/Res...

Limit to 1000 rows

```
1 SELECT contact, COUNT(*) AS total
2 FROM bank_marketing
3 GROUP BY contact;
```

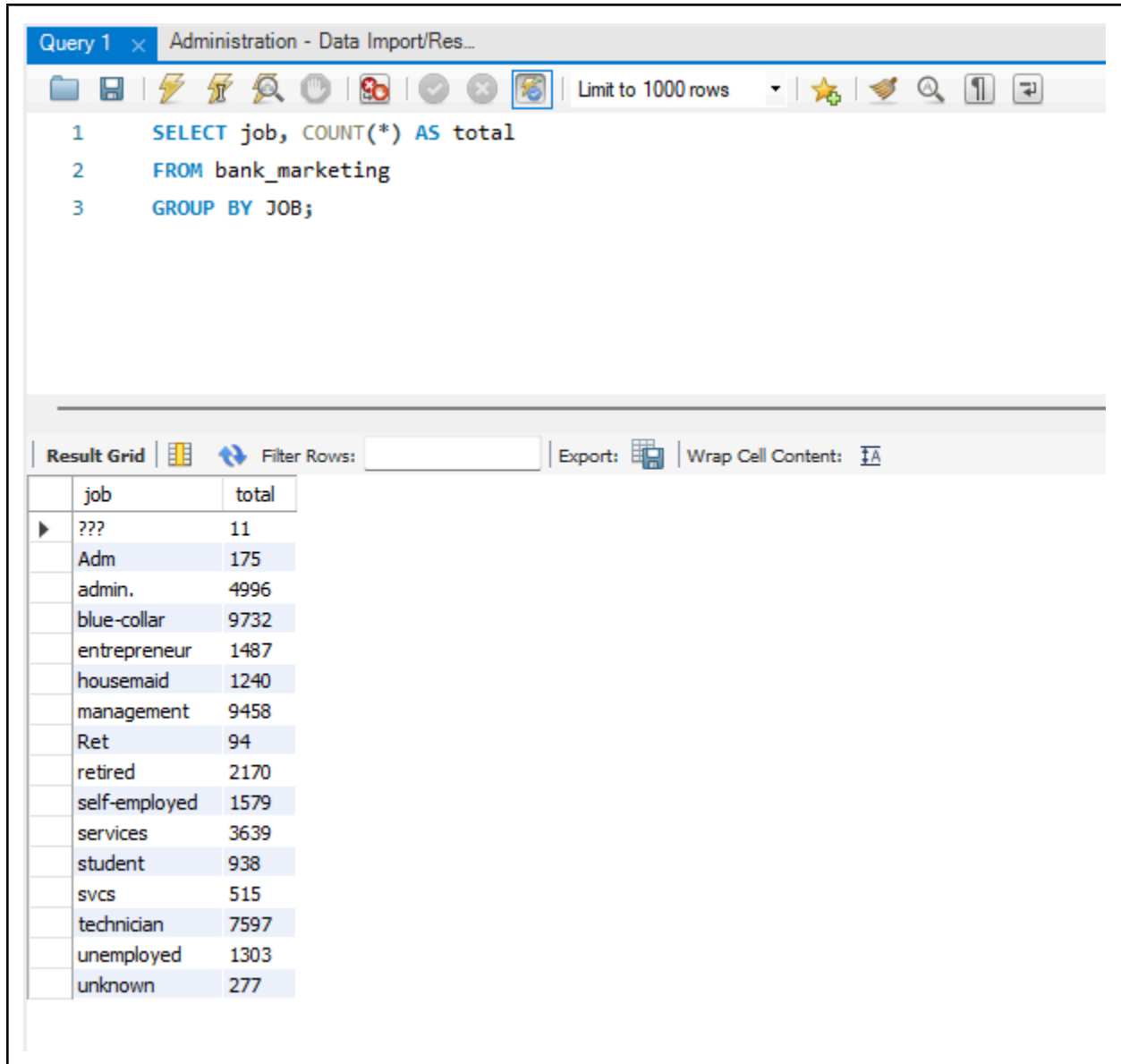
Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

|   | contact   | total |
|---|-----------|-------|
| ▶ | cell.     | 338   |
|   | cellular  | 28947 |
|   | Tel       | 187   |
|   | telephone | 2719  |
|   | unknown   | 13020 |

2.) Based on the summary produced in Reflection item #1, kindly perform data clean-up on the following attributes:

- a. Job
- b. Marital
- c. Education
- d. Contact

3.) Perform another set of data exploration after the data clean-up and complete item #2 in Reflection section.



The screenshot shows a database query tool interface. The top toolbar includes icons for file operations, execution, and a 'Limit to 1000 rows' dropdown. The SQL query editor contains the following code:

```
1 SELECT job, COUNT(*) AS total
2 FROM bank_marketing
3 GROUP BY JOB;
```

Below the query editor is the 'Result Grid' section, which includes a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The results are displayed in a table with two columns: 'job' and 'total'.

| job           | total |
|---------------|-------|
| ???           | 11    |
| Adm           | 175   |
| admin.        | 4996  |
| blue-collar   | 9732  |
| entrepreneur  | 1487  |
| housemaid     | 1240  |
| management    | 9458  |
| Ret           | 94    |
| retired       | 2170  |
| self-employed | 1579  |
| services      | 3639  |
| student       | 938   |
| svcs          | 515   |
| technician    | 7597  |
| unemployed    | 1303  |
| unknown       | 277   |



Query 1 x Administration - Data Import/Res...

Limit to 1000 rows

```
1 SELECT marital, COUNT(*) AS total
2 FROM bank_marketing
3 GROUP BY marital;
```

Result Grid

Filter Rows:

Export: Wrap Cell Content:

|   | marital  | total |
|---|----------|-------|
| ▶ | Div      | 583   |
|   | divorced | 4624  |
|   | married  | 27214 |
|   | single   | 12790 |

Query 1 x Administration - Data Import/Res...

Limit to 1000 rows

```
1 SELECT education, COUNT(*) AS total
2 FROM bank_marketing
3 GROUP BY education;
```

Result Grid

Filter Rows:

Export: Wrap Cell Content:

|   | marital  | total |
|---|----------|-------|
| ▶ | Div      | 583   |
|   | divorced | 4624  |
|   | married  | 27214 |
|   | single   | 12790 |

Query 1

Administration - Data Import/Res...



Limit to 1000 rows



1

SELECT contact, COUNT(\*) AS total

2

FROM bank\_marketing

3

GROUP BY contact;

Result Grid



Filter Rows:

Export: 

Wrap Cell Content: 

|   | contact   | total |
|---|-----------|-------|
| ▶ | cell.     | 338   |
|   | cellular  | 28947 |
|   | Tel       | 187   |
|   | telephone | 2719  |
|   | unknown   | 13020 |

## REFLECTION:

1.) Provide the initial summary for the following attributes (10 pts):

- a. Job – Contains multiple categories (e.g., management, technician, services), but also includes unknown values.
- b. Marital – Categories include married, single, divorced. No unknown values observed.
- c. Education – Includes tertiary, secondary, primary, and 'unknown'.
- d. Contact – Dominated by 'unknown', indicating missing or unclear contact method data.

2.) Provide the final summary for the following attributes (15 pts) :

- a. Job – All unknown entries were replaced with NULL, leaving only valid job categories like technician, management, etc.
- b. Marital – No changes were needed; values like married, single, and divorced were already clean.
- c. Education – unknown entries were cleaned and replaced with NULL, so only valid levels like primary, secondary, and tertiary remain.
- d. Contact – unknown values were replaced with NULL, leaving only valid contact types like cellular or telephone.

3.) Based on the results produced in items #1 and #2, what issues did you encounter during the clean-up process? (5 pts)

- During clean-up, I encountered challenges identifying whether 'unknown' values were truly missing or misclassified, ensuring consistent replacements across attributes, and avoiding accidental data loss.

4.) What are the possible repercussions if data clean-up is neglected during the "Data Preparation" phase of the CRISP-DM framework? (10 pts)

- If data clean-up is neglected, it can lead to inaccurate models, misleading insights, and poor decisions that compromise the entire data mining process. Clean data is essential for building reliable, interpretable, and effective outcomes in CRISP-DM.